

Keywords: study of gas dynamics in galaxy clusters and filaments using cosmological simulations

RESEARCH EXPERIENCES

- **Postdoctoral fellow** Oct 2025 -
Kapteyn Astronomical Institute, University of Groningen Groningen, Netherlands
 - Supervisor: Saleem Zaroubi
 - Research grant funded by the Open University of Israel
- **PhD** Oct 2022 - Oct 2025
Institut d'Astrophysique Spatiale (IAS) Orsay, France
 - Title: "The impact of physical processes in and around galaxy clusters on the estimation of their boundaries and mass: case study of the Virgo cluster numerical replica"
 - Supervisors: Nabila Aghanim & Jenny Sorce
 - Defence date: 2nd of October, 2025

EDUCATION

- **PhD in Astronomy and Astrophysics** 2025
Université Paris-Saclay Orsay, France
- **Magister degree in Fundamental Physics, specialisation in Astrophysics (M2)** 2022
Université Paris-Saclay Orsay, France
- **Bachelor degrees in Physics and Mathematics** 2020
Université de Nantes Nantes, France

PUBLICATIONS

Refereed (rank A) as first author

1. *Correcting the hydrostatic mass for non-thermal gas motions: a comparison of two approaches*
Lebeau, Aghanim, Sorce & Ettori, submitted to A&A
2. *Gas motion in the ICM of the Virgo cluster replica*
Lebeau, Ettori, Aghanim, Sorce & Paste, A&A 707, A336 (2026)
3. *Velocity fields and turbulence from cosmic filaments to galaxy clusters*
Lebeau, Zaroubi, Aghanim, Sorce & Langer, A&A 704, A14 (2025)
4. *Can the splashback radius be an observable boundary of galaxy clusters?*
Lebeau, Ettori, Aghanim & Sorce, A&A 689, A19 (2024)
5. *Mass bias in clusters of galaxies: Projection effects on the case study of Virgo replica*
Lebeau, Sorce, Aghanim, Hernández-Martínez & Dolag, A&A 682, A157 (2024)

Refereed (rank A) as co-author

1. *Simulating the Local Web (SLOW) – V: Thermodynamic properties and evolution of local galaxy clusters*
Hernández-Martínez, Dolag, Steinwandel, Sorce, **Lebeau**, Aghanim & Seidel, A&A 710, A147 (2026)
2. *Simulating the Local Web (SLOW) – II: Properties of local galaxy clusters*
Hernández-Martínez, Dolag, Seidel, Sorce, Aghanim, Pilipenko, Gottlöber, **Lebeau** & Valentini, A&A 687, A253 (2024)

TALKS

Invited talks and seminars

1. **Gas physics in and around galaxy clusters**
Monday Lunch talk of the Kapteyn Astronomical Institute
Feb 2026
Groningen, Netherlands
2. **Can the splashback radius be an observable boundary of galaxy clusters?**
Seminars of the CMB-S4 clusters analysis working group
Mar 2025
online (Univ. of Illinois, USA)
3. **Physics processes of the cosmic gas in galaxy clusters environment**
Day of the astrophysics axis of the Univ. Paris-Saclay Graduate School
Oct 2024
Orsay, France
4. **Physics processes biasing galaxy clusters mass estimation:
case study of the Virgo cluster simulated replica**
INAF-OAS Seminar
Sep 2024
Bologna, Italy

Contributed talks

1. **Simulating small-scale gas dynamics in cosmic filaments**
NAC 2026
May 2026
Egmond aan Zee, Netherlands
2. **Gas motion in the ICM of the Virgo cluster replica**
Ultimate Cluster Cosmology workshop
Dec 2025
Orsay, France
3. **Gas motion in the ICM of the Virgo cluster replica**
EAS annual meeting
Jun 2025
Cork, Ireland
4. **Can the splashback radius be an observable boundary of galaxy clusters?**
Expanding the boundaries of dark matter halos workshop
May 2025
online (Shanghai, China)
5. **Turbulence from cosmic filaments to galaxy clusters**
SNO Ramses days
Nov 2024
Paris, France
6. **Physics processes biasing galaxy clusters mass estimation:
case study of the Virgo cluster simulated replica**
Ultimate cluster cosmology workshop
Oct 2024
Orsay, France
7. **Turbulence in the ICM of the Virgo cluster simulated replica**
EAS annual meeting
Jul 2024
Padova, Italy
8. **Can the splashback radius be an observable boundary of galaxy clusters ?**
GdR Cophy Episode 2
May 2024
Lyon, France
9. **Can the splashback radius be an observable boundary of galaxy clusters ?**
Tuorla-Tartu meeting
May 2024
Turku, Finland
10. **Gas dynamics in the ICM of galaxy clusters: case study of a Virgo replica**
Elbereth Conference
Mar 2024
Paris, France
11. **Mass bias in clusters of galaxies: case study of Virgo CLONE replica**
RAMSES SNO kick-off meeting
Dec 2023
Lyon, France
12. **Biases in the estimation of the hydrostatic mass of the Virgo simulated CLONE**
mm Universe conference
Jun 2023
Grenoble, France
13. **Biases in the estimation of the hydrostatic mass of the Virgo simulated CLONE**
CLUES meeting
Jun 2023
Munich, Germany
14. **Towards bias-free mass calibration of galaxy clusters using constrained
cosmological simulations**
Elbereth Conference
Mar 2023
Paris, France

POSTERS

1. **The WEFT project: simulating small-scale gas dynamics in cosmic filaments**
*Journées SF2A 2026**Jun 2026*
Grenoble, France
2. **Gas motion in the ICM of the Virgo cluster replica**
*Tracing Cosmic Evolution with Galaxy Clusters V Conference**Jul 2025*
Sesto, Italy
3. **Studying physics processes in and around galaxy clusters with cosmological simulations**
*IAS young researchers and engineers day**May 2025*
Orsay, France
4. **Turbulence in galaxy clusters and cosmic filaments**
*IAS young researchers and engineers day**Jun 2024*
Orsay, France
5. **Towards bias-free mass calibration of galaxy clusters using constrained cosmological simulations**
*Colloque Alain Bouyssy**Dec 2023*
Orsay, France
6. **Towards bias-free mass calibration of galaxy clusters using constrained cosmological simulations**
*Journée de l'axe Astro de la Graduate School de Physique de l'Université Paris-Saclay**Oct 2023*
Orsay, France
7. **Towards bias-free mass calibration of galaxy clusters using constrained cosmological simulations**
*IAS young researchers and engineers day**Jun 2023*
Orsay, France
8. **Towards bias-free mass calibration of galaxy clusters using constrained cosmological simulations**
*"Future Cosmology" summer school**Apr 2023*
Cargèse, France

SUPERVISION AND TEACHING

- **Co-Supervision of Léa Gagneux (2nd year Bachelor Degree trainee)**
*3.5 months research internship**Jan-Apr 2025*
- **Co-Supervision of Jade Paste (1st year Master Degree trainee)**
*2 months research internship**May-Jun 2024*
- **Astronomy practical works (15h/year)**
*1st year Master Degree**2022-2024*
- **Electromagnetism courses (21h/year)**
*2nd year Bachelor Degree**2023-2024*
- **Co-supervision of astronomy projects (one week)**
*1 week project with four students of 3rd year Bachelor Degree**2023*

MAIN SKILLS

Computing: Running cosmological simulations on HPC clusters using the AREPO code, analysis of cosmological simulations with Fortran (own RAMSES-related data preparation and map creation codes) and Python (use of scientific (numpy, scipy, astropy,...), visualisation (matplotlib, pyvista,...) and optimisation (numba, jax,...) libraries)
Languages: English (fluent), French (mother tongue)

CONTRIBUTIONS TO THE COMMUNITY

International

- **Referee for "The Open Journal of Astrophysics"***2024*

Local

- **Postdoc representative at the Kapteyn Astronomical Institute***Apr 2026 -*
- **Organisation of Saleem Zaroubi's biweekly group meetings***Nov 2025 -*
- **Member of the LOC for the Ultimate Cluster Cosmology workshop @ IAS***2024*
- **Co-organisation of bimonthly Cosmology team seminars***2024 - 2025*

- Co-organisation of the IAS young researchers and engineers day 2024 & 2025
- Elected as doctoral student representative at the laboratory board 2024 - 2025
- Elected as doctoral student representative at the Paris-Saclay University Physics Graduate School board 2023 - 2025
- Management of the IAS Cosmology team's conference webpage 2022 - 2024

COLLABORATIONS

- Member of the LOCALIZATION project 2022-2025
P.I.s: Nabila Aghanim (IAS, Paris-Saclay University) & Klaus Dolag (LMU, Munich)

GRANTS AND PRIZES

- PhD Prize of the Graduate School of Physics of the Paris-Saclay University (450€) Apr 2026
- Financial support to attend the "Journées de la SF2A 2026" (700€) Mar 2026
- Financial support from doctoral school to participate to "Future Cosmology" summer school (~500€) Apr 2023
- 3-years PhD half-grant from doctoral school 2022 - 2025
"Astronomie & Astrophysique d'Ile-de-France" (~50k€)

PROPOSALS

- Co.I of project Proposal for Tier 0/Tier 1 HPC Access at the Gauss Center for supercomputing 2023
45Mcpu hours obtained on the LRZ supercomputer to run the LOCALIZATION simulation

OUTREACH

- "Introduction to Astrophysics", meeting with 7th grade students Dec 2025
- "Introduction to Astrophysics", meeting with 7th grade students Apr 2025
- "The story of my PhD", ALCOR Astronomy association event Oct 2024
- Conference "Introduction to cosmology" for secondary school students Dec 2023
- Participation to the "Science Festival 2022" at IAS Oct 2022

PERSONAL INTERESTS

- Basketball player and coach in competition
- Guitar

REFERENCES

- Nabila Aghanim: nabila.aghanim@universite-paris-saclay.fr
- Jenny Sorce: jenny.sorce@univ-lille.fr
- Stefano Ettori: stefano.ettori@inaf.it
- Saleem Zaroubi: saleem@astro.rug.nl